

Fourth Year – Second Semester

Data Mining

Lecture (1) – What is Data Mining?

Assist. Prof. Dr. Zainab Namh Sultani
Computer Science Department
College of Science
Al-Nahrain University

Course Objectives

- Data mining involves the intelligent extraction of valuable, novel, and previously unknown insights from large datasets.
- As an interdisciplinary field, it integrates machine learning, statistics and database systems.
- This course introduces the foundational concepts of data mining and provides an in-depth exploration of its algorithms, models, and applications. Topics include data preprocessing, pattern mining, decision trees, cluster analysis, dimensionality reduction, and text mining.
- Through hands-on programming assignments, students will gain practical experience applying these techniques to real-world datasets.

Course Syllabus

- Introduction to Data Mining, Data for Data Mining
- Data Science Methodology
- Data, measurements, and data preprocessing
- Pattern Mining: basic concepts and methods
- Decision Trees
- Cluster analysis: basic concepts and methods
- Dimensionality Reduction
- Text Mining

Course Evaluation

Assessment Type	Marks
Quizzes	5
Seminar	5
Attendance	5
Mid Exam	10
Lab Mid Exam	7
Lab Attendance and Assessment	8
Lab Final Exam	10
Final Exam	50
Total	100

Learning Resources

Required textbooks

Data Mining, Concepts and Techniques, Jiawei Han, Micheline Kamber and Jian Pei, Elsevier, 2023

What is Data Mining?

- We live in a world where vast amounts of data are generated constantly and rapidly. “*We are living in the information age*” is a popular saying; however, *we are actually living in the data age*.
- Terabytes or petabytes of data pour into our computer networks, the World Wide Web (WWW), and various kinds of devices every day from business, news agencies, society, science, engineering, medicine, and almost every other aspect of daily life.
- This explosively growing, widely available, and gigantic body of data makes our time truly the data age. Powerful and versatile tools are badly needed to automatically uncover valuable information from the tremendous amounts of data and to transform such data into organized knowledge. This necessity has led to the birth of data mining.

What is Data Mining?

- Essentially, data mining is **the process of discovering interesting patterns and knowledge from large amounts of data**. The data sources can include databases, data warehouses, the Web, other information repositories, or data that are streamed into the system dynamically.
- The term, data mining, as a vivid view of searching for gold nuggets from data, appeared in 1990s. However, to refer to the mining of gold from rocks or sand, we say gold mining instead of rock or sand mining. Analogously, data mining should have been more appropriately named “knowledge mining from data,” which is unfortunately somewhat long.
- In addition, many other terms have a similar meaning to data mining—for example, *knowledge mining from data*, *KDD (i.e., Knowledge Discovery from Data)*, *pattern discovery*, *knowledge extraction*, *data analytics*, and *information harvesting*.

Example: Data Mining in Amazon

1. Product Recommendation System

When you open Amazon and see: “**Customers who bought this also bought...**”, **Recommended for you**”, “**Frequently bought together**”

This is **Data Mining + Machine Learning**.

How it works: Amazon collects data about: What you search, What you click, What you buy, Time spent on products

It finds **patterns** among millions of users.

If many users who bought *Laptop A* also bought *Mouse B*, Amazon recommends Mouse B to new laptop buyers.

This is called **Association Rule Mining** and **Collaborative Filtering**

Example: Data Mining in Amazon

2. Customer Personalization

- Every user sees a **different homepage**.
- Data mining analyzes: Age group, Location, Past purchases, Browsing history
- Then Amazon predicts:
 - What you might want next
 - What price range you prefer
- This is **Predictive Data Mining (Classification & Regression)**

Data Mining Applications

Netflix / YouTube – Movie & Video Recommendations

- Suggests movies or videos based on what you watched before.
- Analyzes viewing history, likes, watch time.
- Uses **pattern mining and machine learning**.

“If you watch many action movies, Netflix will recommend more action films.”

Data Mining Applications

Facebook / Instagram / TikTok – Social Media Feeds

- Decides which posts or reels appear first.
- Analyzes likes, comments, shares, and watch time.
- Uses **behavior pattern analysis**.

“If you like many cooking videos, your feed becomes full of recipes”.

Data Mining Applications

Spotify – Music Recommendations

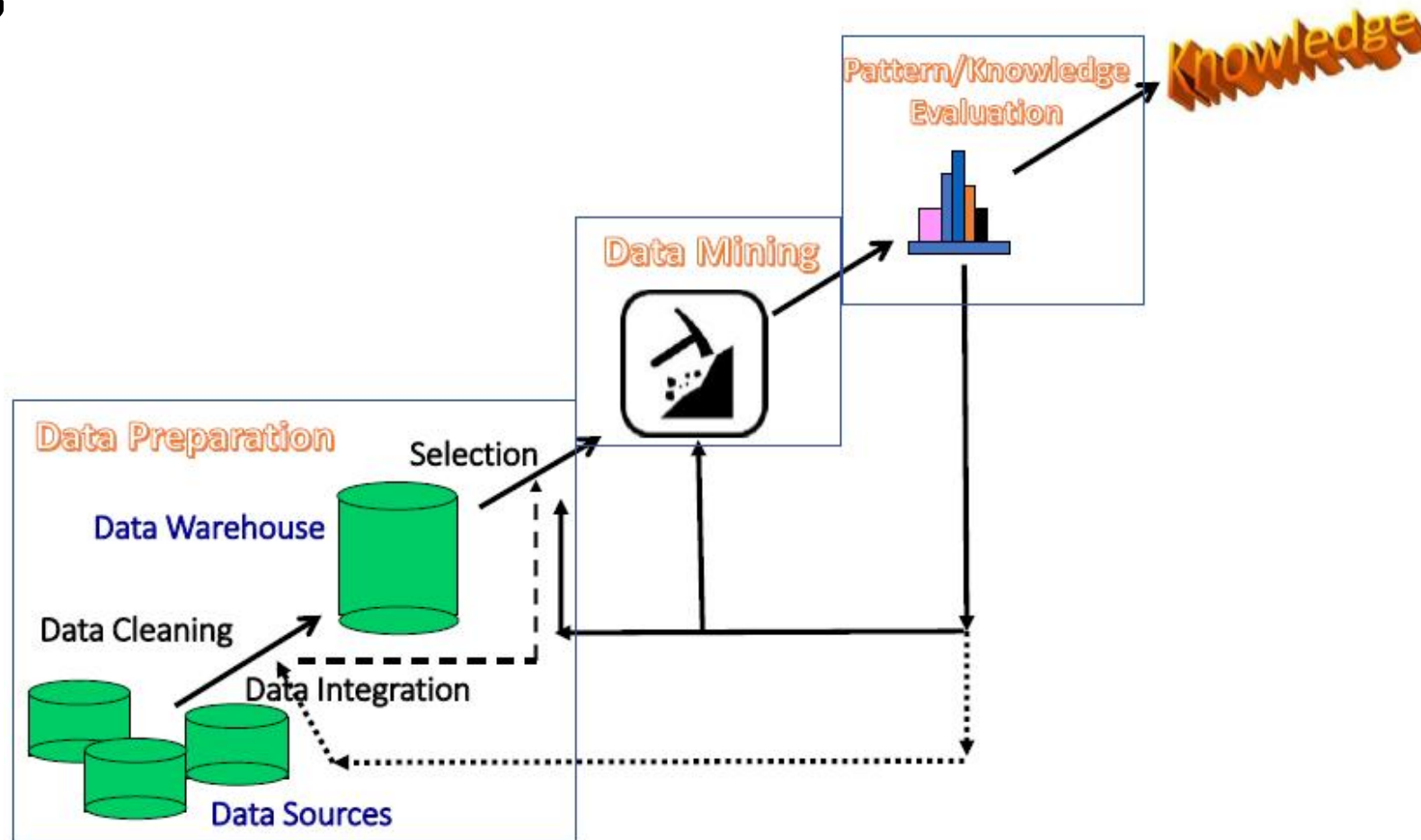
- Suggests playlists like “**Discover Weekly.**”
- Analyzes listening habits, genres, and mood patterns.

Transportation – Uber / Google Maps

- Predicts traffic congestion.
- Suggests fastest routes.
- Uses **real-time streaming data mining.**

Data mining: an essential step in the process of knowledge discovery

- The overall knowledge discovery process is shown in Fig. 1.1 as an iterative sequence of the follo



Data mining: an essential step in knowledge discovery

1. Data preparation

- a. **Data cleaning** (to remove noise and inconsistent data)
- b. **Data integration** (where multiple data sources may be combined)
- c. **Data transformation** (where data are transformed and consolidated into forms appropriate for mining by performing summary or aggregation operations)
- d. **Data selection** (where data relevant to the analysis task are retrieved from the database)

2. Data mining (an essential process where intelligent methods (ML) are applied to extract patterns or construct models)

3. Pattern/model evaluation (to identify the truly interesting patterns or models representing knowledge based on *interestingness measures*)

4. Knowledge presentation (where visualization and knowledge representation techniques are used to present mined knowledge to users)

Diversity of data types for data mining

- As a general technology, data mining can be applied to any kind of data as long as the data are meaningful for a target application.
- Data stored in *relational databases*, *data cubes*, *data matrices*, and many *data warehouses* have uniform, record- or table-like structures, defined by their data dictionaries, with a fixed set of attributes (or fields, columns), each with a fixed set of value ranges and semantic meaning. These data sets are typical examples of highly structured data.
- However, there are also large amounts of unstructured data, such as text data and multimedia (e.g., audio, image, video) data
- Data is filtered, normalized, sampled, transformed in various ways, and eventually used as input to data mining algorithms.
- Up to 80% of the effort in a data mining project is often devoted to data preparation.

Data Mining Functions

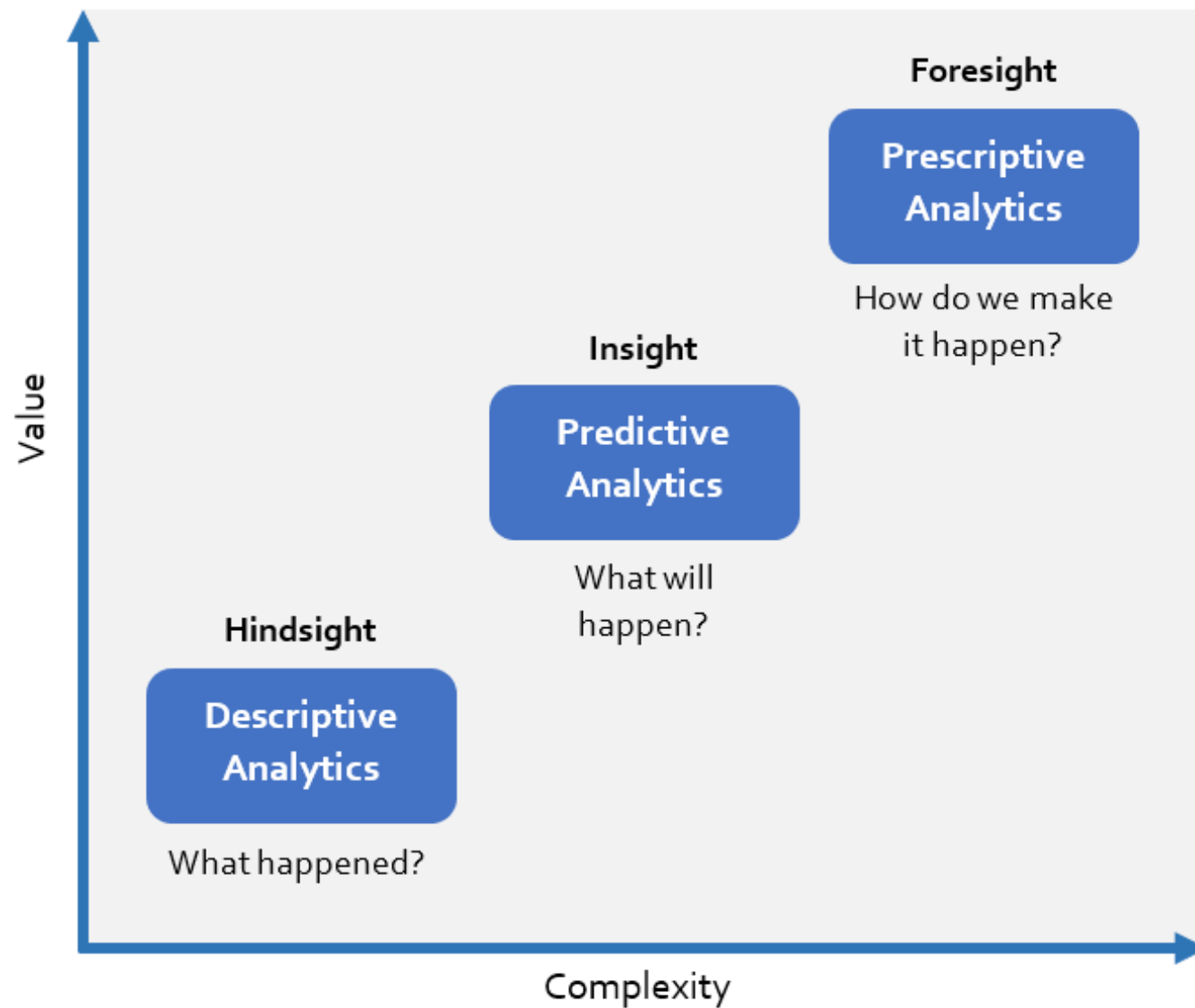
- Data mining functions can be divided into two categories: supervised (directed) and unsupervised (undirected).
- Supervised functions are used to predict a value; they require the specification of a target (known outcome). Targets are either binary attributes indicating yes/no decisions (buy/don't buy) or multi-class targets indicating a preferred alternative (color of sweater, likely salary range, etc.). Naive Bayes for classification is a supervised mining algorithm.
- Unsupervised functions are used to find the intrinsic structure, relations, or affinities in data. Unsupervised mining does not use a target. Clustering algorithms can be used to find naturally occurring groups in data.

Mining various kinds of knowledge

- Different kinds of patterns and knowledge can be uncovered via data mining. In general, data mining tasks can be put into two categories: **descriptive data mining** and **predictive data mining**. Descriptive mining characterizes properties of the interested set of data, whereas predictive mining performs induction on the data set in order to make predictions.
- Predictive data mining constructs one or more models; these models are used to predict outcomes for new data sets. Predictive data mining functions are classification and regression. Naive Bayes is one algorithm used for predictive data mining.
- Descriptive data mining describes a data set in a concise way and presents interesting characteristics of the data. Descriptive data mining functions are clustering, association models, and feature extraction. k-Means clustering is an algorithm used for descriptive data mining.

Data Mining Functions cont. ...

- Different algorithms serve different purposes; each algorithm has advantages and disadvantages. A given algorithm can be used to solve different kinds of problems.
- For example, k-Means clustering is unsupervised data mining; however, if you use k-Means clustering to assign new records to a cluster, it performs predictive data mining. Similarly, decision tree classification is supervised data mining; however, the decision tree rules can be used for descriptive purposes.



Data Mining Functions cont. ...

- **Supervised data mining:** A classic example of supervised learning is classification. The supervision in the learning comes from the labelled examples in the training data set.
 - Classification: Grouping items into discrete classes and predicting which class an item belongs to.
 - Regression: Approximating and forecasting continuous values.
- **Unsupervised data mining:** The learning process is unsupervised since the input examples are not class-labelled.
 - Clustering: Finding natural groupings in the data.
 - Association models: Analyzing "market baskets".
 - Feature extraction: Creating new attributes (features) as a combination of the original attributes.

$age(X, \text{"youth"}) \text{ AND } income(X, \text{"high"}) \longrightarrow class(X, \text{"A"})$
 $age(X, \text{"youth"}) \text{ AND } income(X, \text{"low"}) \longrightarrow class(X, \text{"B"})$
 $age(X, \text{"middle_aged"}) \longrightarrow class(X, \text{"C"})$
 $age(X, \text{"senior"}) \longrightarrow class(X, \text{"C"})$

(a)

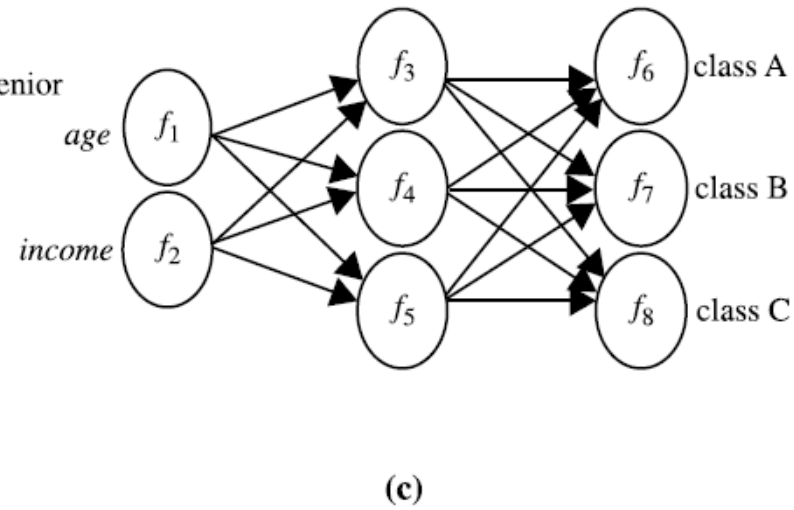
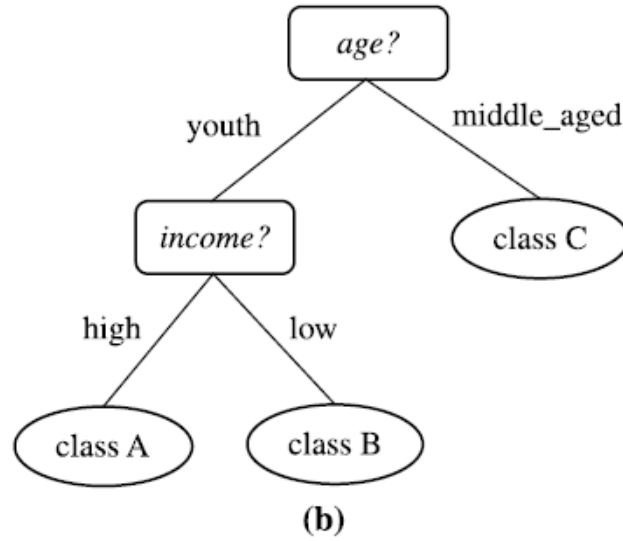


FIGURE 1.2

A classification model can be represented in various forms: (a) IF-THEN rules, (b) a decision tree, or (c) a neural network.

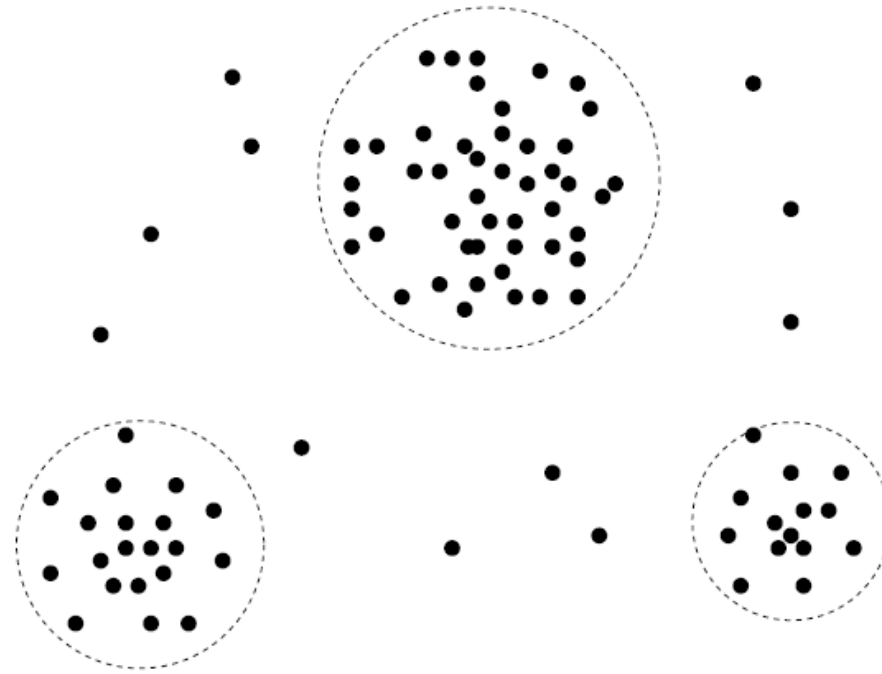


FIGURE 1.3

A 2-D plot of customer data with respect to customer locations in a city, showing three data clusters.

Data for Data Mining

- The data preparation required depends on the type of model that you plan to build and the characteristics of the data. For example, data with attributes that take on a small number of values, that is, that have low cardinality, may not require binning.
- The rows of a data table are often called cases, records, or examples. The columns of the data tables are called attributes or fields; each attribute in a record holds a cell of information.
- Attribute names are constant from record to record; the values in the attributes can vary from record to record. For example, each record may have an attribute labeled "annual income." The value in the annual income attribute can vary from one record to another.

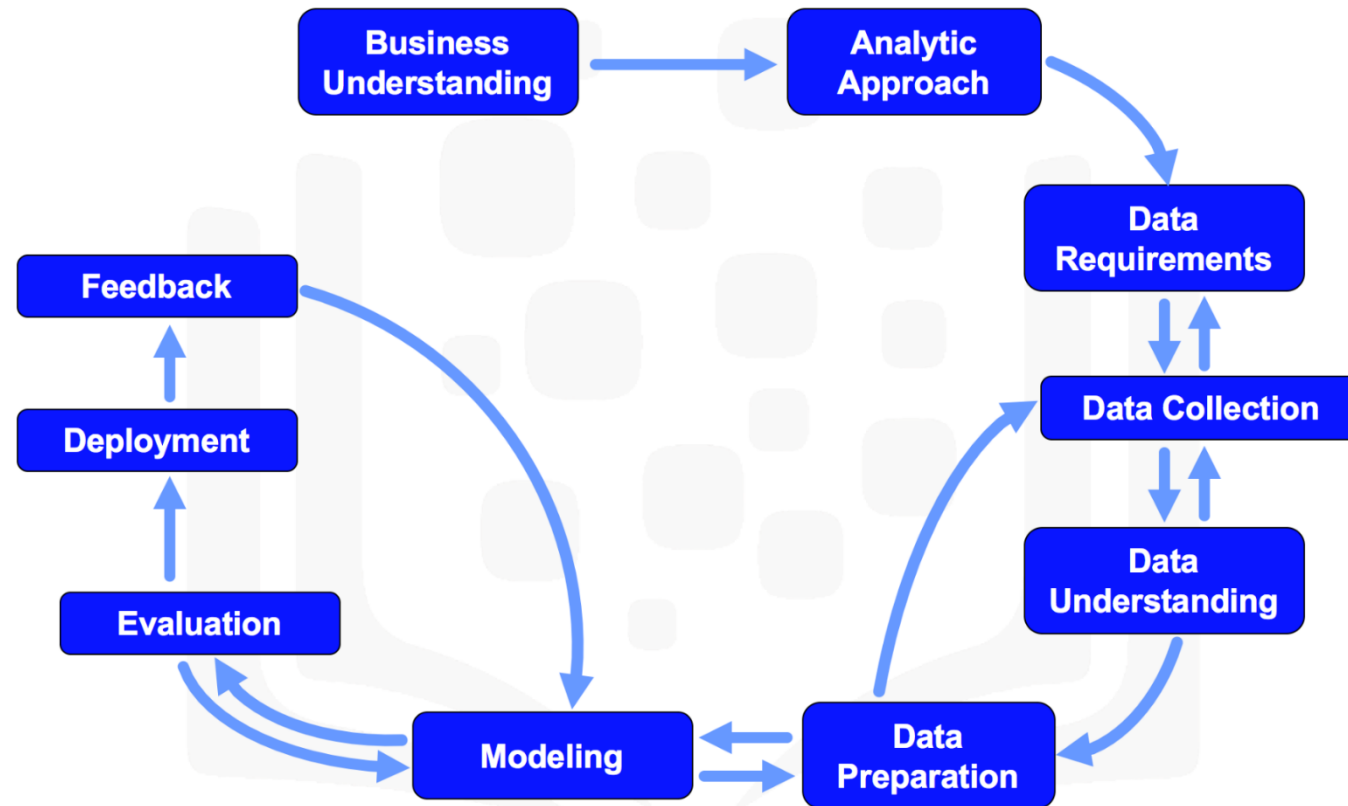
Data mining and other disciplines

- The majority of the real-world data are unstructured, in the form of natural language text, images, or audio-video data. Therefore, natural language processing, computer vision, pattern recognition, audio- video signal processing, and information retrieval will offer critical help at handling such data.
- Actually, handling any special kinds of data will need a lot of domain knowledge to be integrated into the data mining algorithm design.

Data mining and applications

- To demonstrate the importance of applications of data mining, we briefly discuss a few highly successful and popular application examples of data mining: *business intelligence*; *social media*; and *biology, medical science, and health care*.
- **Business intelligence:** Without data mining, many businesses may not be able to perform effective market analysis, compare customer feedback on similar products, discover the strengths and weaknesses of their competitors, retain highly valuable customers.
- **Social media:** Social media mining is to sift through massive amounts of social media data (e.g., on social media usage, online social behaviors, connections between individuals, online shopping behavior, content exchange, etc.) in order to discern patterns and trends.

Data Mining/Science Methodology



Data Mining Process

1. Business Understanding: Understand the project objectives and requirements from a business perspective, and then convert this knowledge into a data mining problem definition and a preliminary plan designed to achieve the objectives.
2. Data Understanding: Start by collecting data, then get familiar with the data, to identify data quality problems, to discover first insights into the data, or to detect interesting subsets to form hypotheses about hidden information.
3. Data Preparation: Includes all activities required to construct the final data set (data that will be fed into the modeling tool) from the initial raw data. Tasks include table, case, and attribute selection as well as transformation and cleaning of data for modeling tools.

Data Mining Process

4.Modeling: Select and apply a variety of modelling techniques, and calibrate tool parameters to optimal values. Some techniques have specific requirements on the form of data. Therefore, stepping back to the data preparation phase is often needed.

5.Evaluation: Thoroughly evaluate the model, and review the steps executed to construct the model, to be certain it properly achieves the business objectives.

6.Deployment: Organize and present the results of data mining. Deployment can be as simple as generating a report or as complex as implementing a repeatable data mining process. Data mining is iterative. A data mining process continues after a solution is deployed.